EFFECTIVE POINT-COUNT THRESHOLDS AND POLYNOMIAL ZETA-RECOVERY BOUNDS FOR ABELIAN VARIETIES OVER FINITE FIELDS

ABSTRACT. Let A be a g-dimensional abelian variety over $\mathbf{F}_q$ with point counts $c_n(A) = \#A(\mathbf{F}_{q^n})$ for $n \geq 1$. Chidambaram and Keller recently proved that $c_1(A), \ldots, c_g(A)$ uniquely determine the zeta function of A whenever $q > (16g^3p(2g))^{2g+2}$, studied the minimal number $N(g)$ of extension point counts required for large q (showing $N(2) = 2$, $N(3) = 2$, and $N(4) = 4$), and posed several questions in Section 7 of their paper concerning $N(5)$ and $N(6)$, the growth of $N(g)$ relative to g, the behavior at genus 8, and whether a polynomial threshold $Q(g)$ suffices for g point counts to determine the zeta function.

In this note we establish the following results:

(1) *Small genera:* Using coupled cyclotomic norm differences, we prove $N(5) = 4$ (for $q \geq 10^5$), $N(6) = 4$ (for $q \geq 10^6$), and $N(7) = 5$ (upper bound for $q \geq 10^6$, together with an explicit ordinary $\mathbf{Z}[q]$-family valid for all prime powers $q \geq 5$).

(2) *Genus 8:* We realize the perturbation direction noted in Remark 7.10 of Chidambaram and Keller by an explicit ordinary $\mathbf{Z}[q]$-family valid for all prime powers $q \geq 4$, proving $N(8) \geq 5$. Conversely, for $N = 5$ we prove that every collision at $q \geq 10^8$ takes the single-parameter form $P_B(T) - P_A(T) = m_4T^4(T^4 - 1)(T^4 - q^4)$ with $1 \leq |m_4| \leq 512$, satisfies $a_2 = a_1q + a_1^2$ and $a_1 \in \{-2, -1, 0, 1\}$ for all sufficiently large q, and admits no $\mathbf{Z}[q]$-family of reciprocal q-Weil polynomials; the exact value $N(8) \in \{5, 6, 7, 8\}$ for sporadic prime powers remains open.

(3) *Higher genera:* Combining Chebyshev base polynomials with totient summation, we prove that $N(g) \geq \left(\frac{\pi}{\sqrt{6}} - o(1)\right)\sqrt{g}$ as $g \to \infty$, and construct explicit ordinary $\mathbf{Z}[q]$-families via lattice reduction establishing $N(15) \geq 7$, $N(24) \geq 8$, and $N(32) \geq 9$.

(4) *Polynomial threshold $Q(g)$:* Refining the inductive error analysis in Section 5 of Chidambaram and Keller via a re-centered geometric envelope $B_j^* = 4gq^{-j/2-(g+1)/4}(10g)^{j-(g+1)/2}$ and the exponential generating function for elementary symmetric polynomials, we prove that $c_1(A), \ldots, c_g(A)$ uniquely determine the zeta function of A whenever $q > (10g)^6 = 10^6g^6$ for all $g \geq 2$.

1. Introduction and Cyclotomic Norm Setup

Let A and B be g-dimensional abelian varieties over a finite field $\mathbf{F}_q$, with Frobenius characteristic polynomials

$$P_A(T) = T^{2g} + a_1T^{2g-1} + \cdots + a_gT^g + qa_{g-1}T^{g-1} + \cdots + q^g = \prod_{j=1}^{2g}(T - \alpha_j) \in \mathbf{Z}[T]$$

and point counts $c_n(A) = \#A(\mathbf{F}_{q^n}) = \prod_{j=1}^{2g}(1 - \alpha_j^n)$. By Tate's isogeny theorem [Tat66], $P_A = P_B$ if and only if A and B are $\mathbf{F}_q$-isogenous. Chidambaram and Keller [CK26] define $N(g)$ as the minimal positive integer N for which there exists a threshold $q_0(g)$ such that for every prime power $q \geq q_0(g)$ and all g-dimensional abelian varieties $A, B/\mathbf{F}_q$, equality $c_n(A) = c_n(B)$ for $1 \leq n \leq N$ implies $P_A = P_B$. In [CK26, Theorem 1.1] they establish $N(g) \leq g$ via an explicit iterative recovery algorithm running in $\widetilde{O}(g^4 \log q)$ bit operations [CK26b] for $q > Q(g) = (16g^3p(2g))^{2g+2}$, and in [CK26, Section 7] they prove $N(2) = 2$, $N(3) = 2$ ($q \geq 16$), and $N(4) = 4$ (with an explicit lower-bound family for $q \geq 8$), while posing Questions 7.8 and 7.9 and Remarks 7.10 and 7.12 on higher genera and polynomial thresholds.

We begin by setting up the real-trace coordinates and cyclotomic norm identities used in Sections 2–4. Write $P_A(T) = T^g h_A(T + q/T)$, where

$$h_A(X) = X^g - b_1 X^{g-1} + b_2 X^{g-2} - \cdots + (-1)^g b_g \in \mathbf{Z}[X]$$

is the real Frobenius trace polynomial, whose roots $x_j = \alpha_j + q/\alpha_j \in [-2\sqrt{q}, 2\sqrt{q}]$ $(1 \le j \le g)$ satisfy the elementary symmetric Weil bounds

$$|b_k| \le B_{g,k}\, q^{k/2}, \qquad B_{g,k} \coloneqq \binom{g}{k} 2^k \qquad (1 \le k \le g). \tag{1}$$

For each integer $d \ge 1$, let $\Phi_d(T)$ denote the d-th cyclotomic polynomial and define the d-th cyclotomic norm

$$F_d(A) \coloneqq \prod_{\Phi_d(\zeta)=0} P_A(\zeta) \in \mathbf{Z}.$$

Since $T^n - 1 = \prod_{d|n} \Phi_d(T)$ and $\deg P_A = 2g$ is even, we have $c_n(A) = \prod_{d|n} F_d(A) > 0$. Moreover, $F_1(A) = P_A(1) = c_1(A) > 0$, $F_2(A) = P_A(-1) = c_2(A)/c_1(A) > 0$, and for $d \ge 3$ every root ζ of Φ_d is non-real on the unit circle, so $P_A(\zeta)P_A(\bar{\zeta}) = |P_A(\zeta)|^2 > 0$ gives $F_d(A) > 0$. By Möbius inversion in $\mathbf{Q}^\times$,

$$c_n(A) = c_n(B) \quad (1 \le n \le N) \qquad \Longleftrightarrow \qquad F_d(A) = F_d(B) \quad (1 \le d \le N). \tag{2}$$

Put $u \coloneqq q + 1$ and $v \coloneqq q - 1$. Using $P_A(\zeta) = \zeta^g h_A(\zeta + q\zeta^{-1})$, one obtains:

- $F_1(A) = h_A(u)$ and $F_2(A) = (-1)^g h_A(-u)$;
- $F_4(A) = P_A(i)P_A(-i) = |h_A(-iv)|^2 = \big(\Re h_A(-iv)\big)^2 + \big(\Im h_A(-iv)\big)^2$;
- $F_3(A) = N_{\mathbf{Q}(\omega)/\mathbf{Q}}(P_A(\omega))$ where $\omega^2 + \omega + 1 = 0$ and $\omega + q\omega^{-1} = -q + (1 - q)\omega$;
- for every $d \ge 3$ ($\varphi(d)$ even), defining the monic trace-cyclotomic polynomial

$$\Psi_d(X, q) \coloneqq \mathrm{Res}_z(\Phi_d(z),\, z^2 - Xz + q) = \prod_{\Phi_d(\zeta)=0} (X - (\zeta + q\zeta^{-1})) \in \mathbf{Z}[q][X] \tag{3}$$

of degree $\varphi(d)$ (together with $\Psi_1(X, q) \coloneqq X - u$ and $\Psi_2(X, q) \coloneqq X + u$), pairing conjugate roots ζ, ζ^{-1} gives $\prod_{\Phi_d(\zeta)=0} \zeta^g = 1$ and hence $F_d(A) = \mathrm{Res}_X(\Psi_d(X, q), h_A(X))$ for all $d \ge 3$.

For two g-dimensional abelian varieties $A, B/\mathbf{F}_q$, write $d_k \coloneqq b_k(B) - b_k(A)$ $(1 \le k \le g)$, so $|d_k| \le 2B_{g,k}q^{k/2}$. Three divisibility relations hold over $\mathbf{Z}$:

(1) $F_1(B) = F_1(A)$ and $F_2(B) = F_2(A)$ hold if and only if $h_B(u) = h_A(u)$ and $h_B(-u) = h_A(-u)$, i.e. if and only if $X^2 - u^2$ divides $h_B(X) - h_A(X)$ in $\mathbf{Z}[X]$.

(2) Whenever $h_B(X) - h_A(X) = (X^2 - u^2)Q(X)$ with $Q \in \mathbf{Z}[X]$, evaluating at $X = -iv$ uses $(-iv)^2 - u^2 = -2(q^2 + 1)$ to give $h_B(-iv) = h_A(-iv) - 2(q^2 + 1)Q(-iv)$ in $\mathbf{Z}[i]$, whence

$$F_4(B) - F_4(A) = -4(q^2 + 1)\,\Re\big(\overline{h_A(-iv)}\, Q(-iv)\big) + 4(q^2 + 1)^2 |Q(-iv)|^2$$

is divisible by $4(q^2 + 1)$ identically, with primitive polynomial quotient

$$E_4 \coloneqq \frac{F_4(B) - F_4(A)}{4(q^2 + 1)}.$$

(3) For every prime ℓ, since $\zeta_\ell \equiv 1 \pmod{1 - \zeta_\ell}$ and $F_1(B) = F_1(A)$, one has $P_B(\zeta_\ell) \equiv P_A(\zeta_\ell) \pmod{1 - \zeta_\ell}$ in $\mathbf{Z}[\zeta_\ell]$; taking norms to $\mathbf{Q}$ shows that ℓ divides $F_\ell(B) - F_\ell(A)$ identically. In particular we set

$$E_3 \coloneqq \frac{F_3(B) - F_3(A)}{3}, \qquad E_5 \coloneqq \frac{F_5(B) - F_5(A)}{5}.$$

Ancillary files. Python and C scripts carrying out the symbolic resultant computations, rational dominance bounds, Sturm root counts, and modular searches in Sections 2–4, together with the coefficient data for Theorem 4.2, accompany this note.

2. Determination of $N(5) = 4$, $N(6) = 4$, and $N(7) = 5$

In [CK26, Example 7.7], Chidambaram and Keller rule out $\mathbf{Z}[q]$-families colliding on $c_1, \dots, c_4$ for $g = 5, 6$, and ask in [CK26, Question 7.8] whether $N(5) = 4$ and $N(6) = 4$ hold across all abelian varieties over $\mathbf{F}_q$ for sufficiently large q. Because collisions at individual prime powers need not arise from polynomial families over $\mathbf{Z}[q]$, proving $N(g) \le N$ requires an archimedean elimination valid for arbitrary integer tuples $(b_1, \dots, b_g)$ satisfying (1). Coupling E_4 with E_3 (and with E_5 at $g = 7$) yields explicit effective thresholds.

Theorem 2.1 ($g = 5$, addressing [CK26, Question 7.8])**.** *For every prime power $q \ge 10^5$ and any 5-dimensional abelian varieties $A, B/\mathbf{F}_q$, if $c_n(A) = c_n(B)$ for $1 \le n \le 4$, then $P_A = P_B$. Combined with $N(5) \ge N(4) = 4$ [CK26, Section 7], we have $N(5) = 4$.*

Proof. Suppose $F_d(A) = F_d(B)$ for $1 \le d \le 4$ and $q \ge 10^5$. Equating the odd and even parts of $h_B(\pm u) = h_A(\pm u)$ gives

$$d_5 + u^2 d_3 + u^4 d_1 = 0, \qquad d_4 + u^2 d_2 = 0.$$

By (1) at $g = 5$, $|d_3| \le 160q^{3/2}$ and $|d_5| \le 64q^{5/2}$. For $q \ge 10^5$,

$$|d_1| = \frac{|u^2 d_3 + d_5|}{u^4} \le \frac{160q^{3/2}(q+1)^2 + 64q^{5/2}}{(q+1)^4} < 1,$$

and since $d_1 \in \mathbf{Z}$, we obtain $d_1 = 0$. Consequently $d_4 = -u^2 d_2$ and $d_5 = -u^2 d_3$, which refine the Weil bounds on $d_2, d_3 \in \mathbf{Z}$ to

$$|d_2| = \frac{|d_4|}{u^2} \le \frac{160q^2}{(q+1)^2} < 160, \qquad |d_3| = \frac{|d_5|}{u^2} \le \frac{64q^{5/2}}{(q+1)^2} < 64q^{1/2}.$$

Expanding $E_4 = (F_4(B) - F_4(A))/(4(q^2+1)) = 0$ in $\mathbf{Z}[q, d_2, d_3, b_1, \dots, b_5]$, its unique highest-weight monomial is $-q^6 d_2$. If $|d_2| \ge 1$, dividing every remaining monomial of E_4 by $|d_2|$ and bounding the variables via $|d_2| \le 160$, $|d_3| \le 64q^{1/2}$, and $|b_k| \le B_{5,k} q^{k/2}$ ($1 \le k \le 5$), every remaining term has q-exponent at most $11/2$, and for all $q \ge 10^5$ their absolute sum is bounded by $\frac{1}{145.55} q^6 |d_2| < q^6 |d_2|$, contradicting $E_4 = 0$. Hence $d_2 = d_4 = 0$.

With $d_1 = d_2 = d_4 = 0$ and $d_5 = -u^2 d_3$, the unique highest-weight monomial of $E_3 = (F_3(B) - F_3(A))/3 = 0$ is $-q^7 d_3$. If $|d_3| \ge 1$, bounding all other terms via $|d_3| \le 64q^{1/2}$ and $|b_k| \le B_{5,k} q^{k/2}$ shows that for all $q \ge 10^5$ their absolute sum is at most $\frac{1}{2200.45} q^7 |d_3| < q^7 |d_3|$, contradicting $E_3 = 0$. Thus $d_3 = d_5 = 0$ and $P_A = P_B$. □

Theorem 2.2 ($g = 6$, addressing [CK26, Question 7.8])**.** *For every prime power $q \ge 10^6$ and any 6-dimensional abelian varieties $A, B/\mathbf{F}_q$, if $c_n(A) = c_n(B)$ for $1 \le n \le 4$, then $P_A = P_B$. Combined with $N(6) \ge N(4) = 4$, we have $N(6) = 4$.*

Proof. Suppose $F_d(A) = F_d(B)$ for $1 \le d \le 4$ and $q \ge 10^6$. From $h_B(\pm u) = h_A(\pm u)$, the odd relation $d_5 + u^2 d_3 + u^4 d_1 = 0$ with $|d_3| \le 320q^{3/2}$ and $|d_5| \le 384q^{5/2}$ gives $|d_1| < 1$ for $q \ge 10^6$, so $d_1 = 0$ and $d_5 = -u^2 d_3$. Parameterizing the even relation $d_6 + u^2 d_4 + u^4 d_2 = 0$ over $\mathbf{Z}$ by $m_4 := d_4 + u^2 d_2 \in \mathbf{Z}$ gives

$$d_4 = m_4 - u^2 d_2, \qquad d_5 = -u^2 d_3, \qquad d_6 = -u^2 m_4.$$

Using (1) at $g = 6$ ($2B_{6,4} = 480$, $2B_{6,5} = 384$, $2B_{6,6} = 128$), successive division by $u^2 = (q+1)^2$ yields, for every $q \ge 1$,

$$|m_4| = \frac{|d_6|}{u^2} < 128q, \qquad |d_3| = \frac{|d_5|}{u^2} < 384q^{1/2}, \qquad |d_2| \le \frac{|d_4| + |m_4|}{u^2} < \frac{480q^2 + 128q}{(q+1)^2} < 480.$$

In $E_4 = 0$, the unique highest-weight monomial is $-q^8 d_2$; for $q \ge 10^6$, assuming $|d_2| \ge 1$ bounds the sum of all remaining terms by $\frac{1}{208.16} q^8 |d_2| < q^8 |d_2|$, forcing $d_2 = 0$.

Once $d_2 = 0$, the remaining parameters (d_3, m_4) are coupled across $E_4|_{d_2=0} = 0$ and $E_3|_{d_2=0} = 0$:

- If $d_3 = 0$, then $E_4|_{d_2=d_3=0} = 0$ has leading monomial $+q^6 m_4$, which dominates all lower-order terms for $q \geq 10^6$ unless $m_4 = 0$.
- Suppose $|d_3| \geq 1$. Isolating $+q^6 m_4$ in $E_4|_{d_2=0} = 0$ against the leading d_3-term $-b_1 q^6 d_3$ (with $|b_1| \leq 12q^{1/2}$) and summing the rational bounds on all remaining terms at $q_0 = 10^6$ gives

$$|m_4| \leq 12.01\, q^{1/2}\, |d_3|.$$

Substituting this bound into $E_3|_{d_2=0} = 0$, whose leading monomial is $-q^9 d_3$, every m_4-bearing term (starting with $+q^7 m_4$, of size at most $12.01\, q^{15/2}|d_3|$) is strictly dominated: for $q \geq 10^6$, the non-leading sum is at most $\frac{1}{81.89}\, q^9 |d_3| < q^9 |d_3|$.

Hence $d_3 = 0$, and therefore $m_4 = 0$, giving $P_A = P_B$. □

At genus 7, four extension point counts no longer determine the zeta function: choosing a degree-2 trace perturbation divisible by $X^2 - (q+1)^2$ automatically preserves F_1 and F_2, leaving enough coefficients in h_A to match F_3 and F_4 simultaneously inside the ordinary q-Weil locus. Including c_5, however, restores uniqueness:

Theorem 2.3 ($g = 7$)**.** *We have $N(7) = 5$. Specifically:*

(1) (Ordinary lower bound for all $q \geq 5$) *Define $h_A, h_B \in \mathbf{Z}[q][X]$ by*

$$\begin{aligned} h_A(X) &= X^7 - (7q+1)X^5 + X^4 + (14q^2+6q)X^3 + (q^2-3q+2)X^2 \\ &\quad - (7q^3+7q^2-1)X - (3q^3-3q^2+5q-1), \\ h_B(X) &= h_A(X) - 2q\big(X^2 - (q+1)^2\big), \end{aligned}$$

and set $P_A(T) = T^7 h_A(T + q/T)$, $P_B(T) = T^7 h_B(T + q/T)$. For every prime power $q \geq 5$, P_A and P_B are distinct ordinary q-Weil polynomials realized by non-isogenous 7-dimensional ordinary abelian varieties $A, B/\mathbf{F}_q$ satisfying $c_n(A) = c_n(B)$ for $1 \leq n \leq 4$ and $c_5(A) \neq c_5(B)$.

(2) (Effective upper bound) *For every prime power $q \geq 10^6$ and any 7-dimensional abelian varieties $A, B/\mathbf{F}_q$, equality $c_n(A) = c_n(B)$ for $1 \leq n \leq 5$ implies $P_A = P_B$.*

Proof. (1) Direct resultant evaluation in $\mathbf{Z}[q]$ gives $F_d(B) - F_d(A) = 0$ for $d \in \{1,2,3,4\}$, while $F_5(B) - F_5(A) = -10q(8q^{11} + \dots) \neq 0$ has no real roots on $[5, \infty)$. Moreover, $a_7(A) \equiv a_7(B) \equiv 1 \pmod q$, so $\gcd(a_7, q) = 1$ in every characteristic. Writing $u_s := q^{-1/2} \in (0, 9/20]$ for $q \geq 5$ (as $(20/9)^2 = 400/81 < 5$) and setting $H_\bullet(x, u_s) := u_s^7 h_\bullet(x/u_s)$, at the eight rational nodes

$$x \in \{-2,\ -\tfrac{9}{5},\ -\tfrac{6}{5},\ -\tfrac{2}{5},\ \tfrac{2}{5},\ \tfrac{6}{5},\ \tfrac{9}{5},\ 2\}$$

both $H_A(x, 0)$ and $H_B(x, 0)$ equal $2T_7(x/2) = x^7 - 7x^5 + 14x^3 - 7x$, which alternates strictly in sign $(-,+,-,+,-,+,-,+)$, and Sturm root counts confirm that none of the sixteen polynomials $H_A(x_m, u_s), H_B(x_m, u_s) \in \mathbf{Q}[u_s]$ vanishes on $(0, 9/20]$. By the intermediate value theorem, for every $q \geq 5$ both h_A and h_B have 7 distinct real roots in $(-2\sqrt{q}, 2\sqrt{q})$ (the threshold $q \geq 5$ is sharp for this pair, as h_B has only 5 real roots in $(-2\sqrt{q}, 2\sqrt{q})$ at $q = 4$). Thus all roots of P_A and P_B lie on $|T| = \sqrt{q}$ with no real roots, and since $\gcd(a_7, q) = 1$, their p-adic Newton polygons have slopes 0 and 1 (each of multiplicity 7), forcing all local invariants in the Brauer group to vanish. By the Honda–Tate theorem [Hon68, Tat71] and the ordinary Weil polynomial criterion [Wat69, §7], [How95, §3], P_A and P_B are realized by non-isogenous 7-dimensional ordinary abelian varieties over $\mathbf{F}_q$.

(2) Suppose $F_d(A) = F_d(B)$ for $1 \leq d \leq 5$ and $q \geq 10^6$. From $h_B(\pm u) = h_A(\pm u)$, the odd relation $d_7 + u^2 d_5 + u^4 d_3 + u^6 d_1 = 0$ with $2B_{7,3} = 560$, $2B_{7,5} = 1344$, $2B_{7,7} = 256$ gives $|d_1| < 1$ for $q \geq 10^6$, hence $d_1 = 0$. Setting $m_4 := d_4 + u^2 d_2 \in \mathbf{Z}$ and $m_5 := d_5 + u^2 d_3 \in \mathbf{Z}$ gives

$$d_4 = m_4 - u^2 d_2, \qquad d_5 = m_5 - u^2 d_3, \qquad d_6 = -u^2 m_4, \qquad d_7 = -u^2 m_5.$$

Dividing successively by $u^2 = (q+1)^2$ gives, for all $q \geq 1$ ($2B_{7,4} = 1120$, $2B_{7,6} = 896$),

$$|m_4| < 896q, \qquad |m_5| < 256q^{3/2}, \qquad |d_3| < 1344q^{1/2}, \qquad |d_2| < 1120.$$

We eliminate (d_2, d_3, m_4, m_5) sequentially for $q \geq 10^6$:

- *Step 1 ($d_2 = 0$):* In $E_4 = 0$, the leading monomial $-q^{10}d_2$ dominates the sum of all remaining terms by a factor of 50.49, forcing $d_2 = 0$.
- *Step 2 ($d_3 = 0$):* When $d_2 = 0$ and $|d_3| \geq 1$, isolating $+q^8 m_4$ in $E_4|_{d_2=0} = 0$ gives $|m_4| \leq 17.59\, q^{1/2}\, |d_3|$. Substituting into $E_3|_{d_2=0} = 0$, the leading monomial $-q^{11}d_3$ dominates all remaining terms by a factor of 46.44, forcing $d_3 = 0$.
- *Step 3 ($m_4 = m_5 = 0$):* When $d_2 = d_3 = 0$, if $m_5 = 0$ then $+q^8 m_4$ in $E_4|_{d_2=d_3=0} = 0$ forces $m_4 = 0$. If $|m_5| \geq 1$, isolating $+q^8 m_4$ in $E_4|_{d_2=d_3=0} = 0$ against $-b_1 q^6 m_5$ ($|b_1| \leq 14q^{1/2}$) yields

  $$|m_4| \leq 14.01\, q^{-3/2}\, |m_5| < 3585.$$

  Substituting this bound into $E_5|_{d_2=d_3=0} = 0$, the leading monomial $-q^{23}m_5$ dominates the sum of all remaining terms by a factor of 69.03 for $q \geq 10^6$. Thus $m_5 = 0$ and $m_4 = 0$, proving $P_A = P_B$. □

3. Genus 8: Realization of Remark 7.10 and Elimination of $N = 5$ $\mathbf{Z}[q]$-Families

In [CK26, Remark 7.10], Chidambaram and Keller exhibit a nonzero perturbation direction at $g = 8$ preserving $F_1, \ldots, F_4$ as identities in q, with $\delta_5 = -(q+1)$, $\delta_7 = q^3 + 1$, and $\delta_k = 0$ for $k \in \{1, 2, 3, 4, 6, 8\}$. We first realize this perturbation direction by an explicit ordinary $\mathbf{Z}[q]$-family valid for every prime power $q \geq 4$, and then analyze $N = 5$ collisions at $g = 8$.

Theorem 3.1 (Realization of [CK26, Remark 7.10]: $N(8) \geq 5$ for all $q \geq 4$)**.** *Define $h_A, h_B \in \mathbf{Z}[q][X]$ by*

$$\begin{aligned} h_A(X) &= X^8 - 7qX^6 - X^5 + (15q^2 - q)X^4 + (-q^2 + 6q)X^3 \\ &\quad + (-11q^3 + 17q^2 - 37q - 2)X^2 + (3q^3 - 8q^2 + q)X + (2q^4 - 15q^3 + 40q^2 + 1), \\ h_B(X) &= h_A(X) - (q+1)X(X^2 - (q+1)^2), \end{aligned}$$

and set $P_A(T) = T^8 h_A(T + q/T)$, $P_B(T) = T^8 h_B(T + q/T)$. Then

$$P_B(T) - P_A(T) = -(q+1)(T^{11} + q^3 T^5) + (q^3 + 1)(T^9 + qT^7),$$

matching the perturbation vector of [CK26, Remark 7.10]*, and for every prime power $q \geq 4$ the polynomials P_A, P_B are distinct ordinary q-Weil polynomials realized by non-isogenous 8-dimensional ordinary abelian varieties $A, B/\mathbf{F}_q$ with $c_n(A) = c_n(B)$ for $1 \leq n \leq 4$ and $c_5(A) \neq c_5(B)$. Consequently, $N(8) \geq 5$.*

Proof. Expanding $P_B(T) - P_A(T) = -(q+1)T^5(T^2 + q)((T^2 + q)^2 - (q+1)^2 T^2)$ gives the stated polynomial with $\delta_5 = -(q+1)$ and $\delta_7 = q^3 + 1$. Direct resultant computation in $\mathbf{Z}[q]$ confirms $F_d(B) - F_d(A) = 0$ for $1 \leq d \leq 4$, while $F_5(B) - F_5(A) = -5q(q+1)(q^{13} + \ldots)$ has no real roots on $[4, \infty)$. Moreover, $a_8(A) \equiv a_8(B) \equiv 1 \pmod{q}$, ensuring ordinariness in every characteristic. At the nine rational nodes

$$x \in \{-2,\ -\tfrac{33}{20},\ -\tfrac{27}{20},\ -\tfrac{4}{5},\ 0,\ \tfrac{4}{5},\ \tfrac{27}{20},\ \tfrac{33}{20},\ 2\},$$

both $H_A(x, 0)$ and $H_B(x, 0)$ equal $x^8 - 7x^6 + 15x^4 - 11x^2 + 2$, which alternates strictly in sign $(+, -, +, -, +, -, +, -, +)$. Sturm root counts show $H_A(x_m, u_s) \neq 0$ on $(0, 1/2]$ ($q \geq 4$) and $H_B(x_m, u_s) \neq 0$ on $(0, 1/3]$ ($q \geq 9$), while Sturm counts at the four remaining prime powers $q \in \{4, 5, 7, 8\}$ confirm that $h_B(X)$ has 8 distinct real roots in $(-2\sqrt{q}, 2\sqrt{q})$ there as well (the threshold $q \geq 4$ is sharp for this pair, as h_B has only 4 real roots in $(-2\sqrt{q}, 2\sqrt{q})$ at $q = 3$). Applying Honda–Tate theory [Tat71, How95] as in Theorem 2.3(1) yields the result. □

At $g = 8$ and $N = 5$, the reciprocal polynomial $T^4(T^4-1)(T^4-q^4)$ of degree $12 < 16$ vanishes identically on Φ_1, Φ_2, and Φ_4, so $F_4(B) = F_4(A)$ alone cannot eliminate its coefficient m_4. Nevertheless, combining archimedean elimination on (E_3, E_4, E_5) with Newton-sum Cauchy–Schwarz bounds and modular elimination rules out all $\mathbf{Z}[q]$-families on $c_1, \dots, c_5$:

Proposition 3.2 (Structure of $g = 8, N = 5$ collisions and non-existence of $\mathbf{Z}[q]$-families)**.** *Let $g = 8$.*

(1) (Effective single-parameter reduction and Cauchy–Schwarz bounds) *For every prime power $q \geq 10^8$ and any 8-dimensional abelian varieties $A, B/\mathbf{F}_q$ with $P_A \neq P_B$ and $c_n(A) = c_n(B)$ for $1 \leq n \leq 5$, one has*

$$P_B(T) - P_A(T) = m_4\, T^4(T^4-1)(T^4-q^4) \tag{4}$$

for a nonzero integer $m_4 \in \mathbf{Z}$ satisfying $1 \leq |m_4| \leq 512$ (and $1 \leq |m_4| \leq 520$ for every $q \geq 12$ whenever (4) holds). Moreover, for all sufficiently large q, any such collision satisfies

$$a_2(A) = a_1(A)\, q + a_1(A)^2 \qquad \text{with} \qquad a_1(A) \in \{-2, -1, 0, 1\}.$$

(2) (No $\mathbf{Z}[q]$-family of reciprocal q-Weil polynomials) *There exist no distinct polynomials $P_A, P_B \in \mathbf{Z}[q][T]$ such that for infinitely many prime powers q, P_A and P_B specialize to characteristic polynomials of 8-dimensional abelian varieties over $\mathbf{F}_q$ (or, more generally, to reciprocal q-Weil polynomials of the form $T^8 h(T + q/T)$ with $h \in \mathbf{Z}[X]$ having all roots in $[-2\sqrt{q}, 2\sqrt{q}]$) satisfying $c_n(A) = c_n(B)$ for $1 \leq n \leq 5$.*

Proof. (1) Assume $q \geq 10^8$ and $F_d(A) = F_d(B)$ for $1 \leq d \leq 5$. From $h_B(\pm u) = h_A(\pm u)$, the odd relation $d_7 + u^2 d_5 + u^4 d_3 + u^6 d_1 = 0$ with $2B_{8,3} = 896$, $2B_{8,5} = 3584$, $2B_{8,7} = 2048$ gives $|d_1| < 1$ for $q \geq 10^8$, so $d_1 = 0$. Hence $h_B(X) - h_A(X)$ is divisible by $X^2 - u^2$:

$$h_B(X) - h_A(X) = d_2(X^6 - u^2X^4) + d_3(X^5 - u^2X^3) + m_4(X^4 - u^2X^2) + m_5(X^3 - u^2X) + \mu_6(X^2 - u^2)$$

with $d_2, d_3, m_4, m_5, \mu_6 \in \mathbf{Z}$. Since $d_8 = -u^2\mu_6$, $d_7 = -u^2 m_5$, $d_6 = \mu_6 - u^2 m_4$, $d_5 = m_5 - u^2 d_3$, and $d_4 = m_4 - u^2 d_2$, successive division by $u^2 = (q+1)^2$ using (1) ($2B_{8,4} = 2240$, $2B_{8,5} = 2B_{8,6} = 3584$, $2B_{8,7} = 2048$, $2B_{8,8} = 512$) gives for all $q \geq 1$:

$$|\mu_6| < 512q^2, \quad |m_5| < 2048q^{3/2}, \quad |m_4| < 3584q, \quad |d_3| < 3584q^{1/2}, \quad |d_2| < 2240.$$

We eliminate d_2, d_3, m_5, and $\nu_6 := \mu_6 - v^2 m_4$ at $q_0 = 10^8$:

- *Eliminating d_2 and d_3:* In $E_4 = 0$, the leading monomial $-q^{12}d_2$ dominates the sum of all remaining terms by a factor of 1637.9 for $q \geq 10^8$, forcing $d_2 = 0$. Setting $d_2 = 0$, if $|d_3| \geq 1$ then isolating $+q^{10}m_4$ in $E_4|_{d_2=0} = 0$ bounds $|m_4|/|d_3| = O(q^{1/2})$, under which $-q^{13}d_3$ in $E_3|_{d_2=0} = 0$ dominates all remaining terms by a factor of 440.5; hence $d_3 = 0$. Isolating $+q^{10}m_4$ in $E_4|_{d_2=d_3=0} = 0$ then gives the preliminary constant bound $|m_4| \leq 33280$.
- *Coupled elimination of (m_5, ν_6):* Write $\mu_6 = v^2 m_4 + \nu_6$ ($v = q - 1$). Since $(X^4 - u^2X^2) + v^2(X^2 - u^2) = (X^2 - u^2)(X^2 + v^2)$ vanishes at $X = \pm iv$, the variable m_4 cancels identically in $E_4|_{d_2=d_3=0}$. In the remaining variables (m_5, ν_6), $E_4|_{d_2=d_3=0} = 0$ has leading terms $-q^8\nu_6 + b_1 q^8 m_5 + \dots$; if $m_5 = 0$ this forces $\nu_6 = 0$, while for $|m_5| \geq 1$ substituting $\nu_6 = b_1 m_5 + r_6$ gives $|r_6| \leq 2240.01\, q^{-1/2}|m_5|$ and $|\nu_6| \leq 16.01\, q^{1/2}|m_5|$ for $q \geq 10^8$.
- In $E_5|_{d_2=d_3=0} = 0$, the leading terms are $-q^{27}m_5 + b_1 q^{27} m_4 + \dots$. If $m_4 = 0$, $-q^{27}m_5$ dominates all (m_5, ν_6)-terms by a factor of 623.5, forcing $m_5 = \nu_6 = 0$ (contradicting $P_A \neq P_B$). Hence $|m_4| \geq 1$, and isolating $-q^{27}m_5$ gives $|m_5| \leq 16.04\, q^{1/2}|m_4|$, so $|\nu_6| \leq 256.8\, q|m_4|$. Substituting into $d_8 = -u^2(v^2 m_4 + \nu_6)$ with $|d_8| \leq 512q^4$ yields $|m_4| \leq 512$ for all $q \geq 10^8$ (and $|m_4| \leq 512q^4/(q^2-1)^2 \leq 520$ for all $q \geq 12$ once $\nu_6 = 0$).
- Next, substituting $\nu_6 = b_1 m_5 + r_6$ into $E_3|_{d_2=d_3=0} = 0$ (whose leading b_1-term is $-q^{11}b_1 m_4$) bounds the integer $|b_1| \leq 120$. Feeding $|b_1| \leq 120$ back into $E_5|_{d_2=d_3=0} = 0$ refines $|m_5| \leq 240.4\, |m_4|$ and $|\nu_6| \leq 28901\, |m_4|$. Isolating $c_2 := b_2 + (b_1 + 8)q$ in $E_3|_{d_2=d_3=0} = 0$ (leading

term $-q^{10}c_2m_4$) then gives $|c_2| \le 31001$. Finally, substituting $b_2 = (-b_1 - 8)q + c_2$ into $E_5|_{d_2=d_3=0} = 0$ cancels all $q^{27}m_4$ terms so that terms involving m_4 alone (with $m_5 = \nu_6 = 0$) have q-degree at most 26; for $q \ge 10^8$, $-q^{27}m_5$ dominates the entire remainder by a factor of 4.30, forcing $m_5 = 0$ and hence $\nu_6 = 0$.

With $d_1 = d_2 = d_3 = m_5 = \nu_6 = 0$, we have $h_B(X) - h_A(X) = m_4(X^2 - u^2)(X^2 + v^2)$, which pulls back under $X = T + q/T$ to (4).

Now let q be sufficiently large with (4). Dividing $F_3(B) - F_3(A) = 0$ by $3m_4$ gives

$$\begin{aligned}(5)\quad 0 &= a_1(q^{11}+1) - a_2(q^{10}+q^6+q^4+1) + a_3(q^5+q^4) + a_4(q^8+1) - a_5(q^7+q^4+q^3+1)\\ &\quad + a_6(q^4+q^2) + a_7(q^5+1) - a_8(q^4+1) + q^8 + q^4 + m_4(q^8+q^4+1).\end{aligned}$$

Since $|a_1| = |b_1| \le 120$ and $|a_2 - a_1q| = |c_2| \le 31001$, write $a_2 = a_1q + r_2$ ($r_2 \in \mathbf{Z}$, $|r_2| \le 31001$). Then $a_1q^{11} - a_2q^{10} = -r_2q^{10}$ in (5), giving $a_4 = r_2q^2 + O(q^{3/2})$. Writing $s = \sqrt{q}$ and $a_k = t_ks^k$ for $k \in \{3,5,6,7,8\}$ ($t_k = O(1)$), expanding $(F_5(B) - F_5(A))/(5m_4) = 0$ in powers of s vanishes identically at s^{54} and s^{53} and has s^{52}-coefficient $r_2 - a_1^2$; since $r_2, a_1^2 \in \mathbf{Z}$, this forces $r_2 = a_1^2$ for all sufficiently large q, and the s^{50}-coefficient gives $t_6 = a_1^3 + t_3^2 + o(1)$.

Let $x_1, \dots, x_8 \in [-2,2]$ be the scaled Frobenius traces $(\alpha_j + \bar{\alpha}_j)/\sqrt{q}$ of A, with elementary symmetric polynomials $\tilde{b}_k = q^{-k/2}b_k(A)$ and power sums $p_k = \sum_{j=1}^8 x_j^k$. From $P_A(T) = \sum_{k=0}^8 (-1)^k b_k T^k (T^2+q)^{8-k}$ and the asymptotics $a_1 = O(1)$, $a_2 = a_1q + a_1^2$, $a_4 = a_1^2q^2 + O(q^{3/2})$, $a_6 = (a_1^3 + t_3^2)q^3 + o(q^3)$, one obtains as $q \to \infty$:

$$\tilde{b}_1 = o(1), \qquad \tilde{b}_2 = a_1 - 8 + o(1), \qquad \tilde{b}_3 = -t_3 + o(1), \qquad \tilde{b}_4 = a_1^2 - 6a_1 + 20 + o(1),$$
$$\tilde{b}_5 = 5t_3 - t_5 + o(1), \qquad \tilde{b}_6 = a_1^3 + t_3^2 - 4a_1^2 + 9a_1 - 16 + o(1),$$

and by Newton's identities, $p_2 = 16 - 2a_1 + o(1)$ and $p_4 = 48 - 8a_1 - 2a_1^2 + o(1)$. Applying Cauchy–Schwarz to $(1, \dots, 1)$ and $(x_1^2, \dots, x_8^2)$ gives

$$0 \le 8p_4 - p_2^2 = 128 - 20a_1^2 + o(1),$$

forcing $|a_1| \le 2$ for sufficiently large q. At $a_1 = 2$, Newton's identities give $p_2 = 12 + o(1)$, $p_4 = 24 + o(1)$, and $p_6 = 36 - 3t_3^2 + o(1)$, so Cauchy–Schwarz on $(x_1, \dots, x_8)$ and $(x_1^3, \dots, x_8^3)$ yields

$$0 \le p_2p_6 - p_4^2 = -144 - 36t_3^2 + o(1) \le -144 + o(1),$$

which is impossible for sufficiently large q. Thus $a_1 \in \{-2,-1,0,1\}$.

(2) Note first that every step in the proof of part (1) uses only the trace representation $P_\bullet(T) = T^8h_\bullet(T + q/T)$ with $h_\bullet \in \mathbf{Z}[X]$ having all roots in $[-2\sqrt{q}, 2\sqrt{q}]$, and hence applies directly to arbitrary reciprocal q-Weil polynomials of degree 16. Suppose $P_A, P_B \in \mathbf{Z}[q][T]$ are distinct polynomials specializing for infinitely many prime powers q to reciprocal q-Weil polynomials of degree 16 with $c_n(A) = c_n(B)$ for $1 \le n \le 5$. Write $a_k(A) = \sum_{j=0}^{\lfloor k/2 \rfloor} c_{k,j}q^j$ ($c_{k,j} \in \mathbf{Z}$, 24 integer unknowns). By part (1), for all sufficiently large q in this infinite set, $P_B(T) - P_A(T) = m_4(q)T^4(T^4-1)(T^4-q^4)$ with $1 \le |m_4(q)| \le 512$; since $P_B - P_A \in \mathbf{Z}[q][T]$, $m_4 \in \mathbf{Z} \setminus \{0\}$ is constant with $1 \le |m_4| \le 512$, $a_1 = u_1 \in \{-2,-1,0,1\}$, and $a_2 = u_1q + u_2$ with $u_2 = u_1^2$.

Equating all q-powers to zero in (5) yields 12 independent linear equations over $\mathbf{Z}$. Solving them expresses 12 coefficients linearly in the remaining 12 coefficients and m_4; in particular, the constant-term equation reads

$$u_2 = c_{2,0} = -c_{6,1} + c_{6,2} - c_{7,0} + c_{8,1} - c_{8,4} + \tfrac{1}{2}m_4 + 1,$$

forcing $m_4 = 2k_4$ with $k_4 \in \mathbf{Z}$ and $1 \le |k_4| \le 256$ (or $|k_4| \le 260$ under the $q \ge 12$ bound $|m_4| \le 520$), with every solved $c_{k,j}$ a $\mathbf{Z}$-linear combination of the 12 free $c_{k,j}$'s and k_4. Substituting into $(F_5(B) - F_5(A))/(5m_4) \in \mathbf{Z}[q]$ gives a degree-26 polynomial in q whose q^{26}-coefficient is $u_2 - u_1^2$ and (upon eliminating $c_{8,4}$ via $u_2 = u_1^2$) whose q^{25}-coefficient is $u_1^3 - c_{6,3}$. Setting $u_1 \in \{-2,-1,0,1\}$, $u_2 = u_1^2$, and $c_{6,3} = u_1^3$, the coefficients of q^{24} and q^{23} eliminate $(c_{6,2}, c_{7,2})$ linearly, and after

substituting $c_{7,1} = c_{6,1} + c_{7,0} - W$ (setting $W \coloneqq c_{6,1} + c_{7,0} - c_{7,1}$), the coefficients of q^{22} and q^{21} eliminate $(c_{6,1}, c_{7,0})$ linearly with constant integer pivots dividing 15; the remaining equations are polynomials in $(c_{3,1}, c_{4,1}, c_{5,1}, c_{5,0}, W, k_4)$, with the q^{20}-equation linear in $c_{5,0}$.

For every prime $p \geq 7$, all pivots dividing 15 are units modulo p. Exhaustive modular evaluation over $\mathbf{F}_p^4 \times (\mathbf{F}_p \setminus \{0\})$ (solving linearly for $c_{5,0} \bmod p$ when its q^{20}-coefficient is nonzero, or testing all $c_{5,0} \in \mathbf{F}_p$ when it vanishes) across the four branches $u_1 \in \{-2, -1, 0, 1\}$ finds no solutions with $k_4 \not\equiv 0 \pmod p$ for each $p \in \{7, 11, 13\}$. Consequently $7 \cdot 11 \cdot 13 = 1001 \mid k_4$, contradicting $1 \leq |k_4| \leq 256$. □

Remark 3.3 (Status of $N(8)$). Theorem 3.1 and [CK26, Theorem 1.1] give $5 \leq N(8) \leq 8$, while Proposition 3.2 shows that no $\mathbf{Z}[q]$-family can raise $N(8)$ above 5 and restricts any sporadic prime-power collision on $c_1, \ldots, c_5$ to (4) with $a_2 = a_1 q + a_1^2$ and $a_1 \in \{-2, -1, 0, 1\}$. Determining whether sporadic non-polynomial collisions occur for arbitrarily large q, or whether further archimedean positivity constraints rule out the remaining four values of a_1 to establish $N(8) = 5$, remains open.

4. Higher Genera: Asymptotic $\sqrt{g}$ Lower Bound and Sublinear Families

In [CK26, Question 7.9], Chidambaram and Keller ask how $N(g)$ grows relative to g as g increases, noting that $N(g) = g$ for $g = 1, 2, 4$ whereas $N(3) = 2 < 3$ (and, as Remark 7.2 of [CK26] recalls via DiPippo and Howe [DH98, Theorem 1.1], the volume of the ordinary Weil locus is substantial). Theorems 2.1–2.3 show that strict sublinearity $N(g) < g$ holds at $g = 5, 6, 7$. Conversely, combining scaled Chebyshev polynomials with cyclotomic trace factors yields an $\Omega(\sqrt{g})$ lower bound on $N(g)$, complemented by lattice-reduced ordinary $\mathbf{Z}[q]$-families in moderate genera.

Theorem 4.1 (Asymptotic $\sqrt{g}$ lower bound, addressing [CK26, Question 7.9])**.** *For every integer $N \geq 2$ and every integer $g \geq 2\sum_{d=1}^{N} \varphi(d)$, we have $N(g) \geq N+1$. Consequently, since $\sum_{d=1}^{N} \varphi(d) = \frac{3}{\pi^2} N^2 + O(N \log N)$ [HW08, Theorem 330],*

$$N(g) \geq \left(\frac{\pi}{\sqrt{6}} - o(1) \right) \sqrt{g} \qquad (g \to \infty),$$

where $\pi/\sqrt{6} \approx 1.28255$.

Proof. Since $N(g)$ is non-decreasing in g (taking products with an ordinary elliptic curve over $\mathbf{F}_q$ preserves equality of point counts while incrementing the dimension), it suffices for each $N \geq 2$ to prove $N(g) \geq N + 1$ at the even integer $g = 2D_N \geq 4$, where $D_N \coloneqq \sum_{d=1}^{N} \varphi(d) \geq 2$.

Define $\Lambda_N(X, q) \coloneqq \prod_{d=1}^{N} \Psi_d(X, q) \in \mathbf{Z}[q][X]$ using (3). Here $\Psi_1(X, q)\Psi_2(X, q) = X^2 - (q+1)^2$, and for each $d \geq 3$ ($\varphi(d)$ even), grouping conjugate roots $\{\zeta, \zeta^{-1}\}$ of Φ_d gives

$$\Psi_d(X, q) = \prod_{\substack{\Phi_d(\zeta) = 0 \\ \Im\zeta > 0}} \Big(X^2 - c_\zeta (q+1) X + q^2 + (c_\zeta^2 - 2)q + 1 \Big), \qquad c_\zeta = \zeta + \zeta^{-1}.$$

In every quadratic factor (including $\Psi_1\Psi_2$), the coefficient of X^{2-r} ($r \in \{0, 1, 2\}$) has q-degree at most r, with q^2-coefficient $+1$ (for $d \geq 3$) or -1 (for $\Psi_1\Psi_2$) at $r = 2$, and constant term $+1$ (for $d \geq 3$) or -1 (for $\Psi_1\Psi_2$) at $X = q = 0$. Multiplying over $1 \leq d \leq N$ shows that $\Lambda_N(X, q)$ is monic of X-degree D_N, the coefficient of $X^{D_N - r}$ ($0 \leq r \leq D_N$) has q-degree at most r, and $\Lambda_N(0, q) = -q^{D_N} + \cdots - 1 \equiv -1 \pmod q$. In particular, under $X = x\sqrt{q}$, any monomial $q^j X^{D_N - r}$ of $\Lambda_N(X, q)$ equals $x^{D_N - r} q^{j + (D_N - r)/2}$, where $j + (D_N - r)/2 \leq r + (D_N - r)/2 = (D_N + r)/2 \leq D_N = g/2$, with equality $j = r = D_N$ occurring solely at the leading term $-q^{D_N}$ of $\Lambda_N(0, q)$.

Let $T_g(y) = \cos(g \arccos y)$ be the g-th Chebyshev polynomial of the first kind. Since $g = 2D_N$ is even, $C_g(X, q) \coloneqq 2q^{g/2} T_g(X/(2\sqrt{q})) \in \mathbf{Z}[q][X]$ is monic of degree g with $C_g(0, q) = 2(-q)^{D_N} \equiv 0 \pmod q$. Define

$$h_A(X) \coloneqq C_g(X, q) + q^{g/2} + 1, \qquad h_B(X) \coloneqq h_A(X) + 2\Lambda_N(X, q).$$

We check the required properties:

(1) *Valid q-degrees and ordinariness:* For $0 \le r \le D_N$, the term X^{D_N-r} of $\Lambda_N(X,q)$ sits at index $k = g-(D_N-r) = D_N+r \ge 2r$ in $h_B(X)$, so its q-degree $j \le r \le \lfloor k/2 \rfloor$ satisfies the Weil polynomial degree bound. Moreover, $a_g(A) = h_A(0) \equiv 1 \pmod q$ and $a_g(B) = h_B(0) \equiv 1+2(-1) = -1 \pmod q$, so $\gcd(a_g, q) = 1$ in every characteristic.

(2) *Cyclotomic agreement:* Since $\Psi_d(X,q) \mid (h_B(X) - h_A(X))$ in $\mathbf{Z}[q][X]$ for $1 \le d \le N$, we have $F_d(B) = F_d(A)$ for $1 \le d \le N$ and hence $c_n(A) = c_n(B)$ for $1 \le n \le N$, while $h_B \ne h_A$ implies $P_B \ne P_A$.

(3) *Real roots in $(-2\sqrt{q}, 2\sqrt{q})$ for $q \ge q_0(N)$:* Substituting $X = x\sqrt{q}$ ($x \in [-2,2]$), every term of $2\Lambda_N(x\sqrt{q}, q)$ other than $-2q^{g/2}$ has q-exponent at most $(g-1)/2$, so uniformly on $[-2,2]$ as $q \to \infty$:

$$q^{-g/2} h_A(x\sqrt{q}) = 2T_g(x/2) + 1 + O_N(q^{-g/2}), \qquad q^{-g/2} h_B(x\sqrt{q}) = 2T_g(x/2) - 1 + O_N(q^{-1/2}).$$

At the $g+1$ Chebyshev extrema $x_m = 2\cos(m\pi/g) \in [-2,2]$ ($0 \le m \le g$), we have $2T_g(x_m/2) = 2(-1)^m$, so $2T_g(x_m/2)+1 \in \{+3,-1\}$ and $2T_g(x_m/2)-1 \in \{+1,-3\}$ both alternate strictly in sign across $m = 0,1,\dots,g$ with absolute value at least 1. Since the $O_N(q^{-1/2})$ coefficient bound is explicit, there exists an explicit threshold $q_0(N)$ such that for every prime power $q \ge q_0(N)$, both h_A and h_B change sign strictly between consecutive nodes x_m and therefore have g distinct real roots in $(-2\sqrt{q}, 2\sqrt{q})$.

By Honda–Tate theory [Tat71, How95], P_A and P_B arise from non-isogenous g-dimensional ordinary abelian varieties over $\mathbf{F}_q$ for all prime powers $q \ge q_0(N)$, proving $N(g) \ge N+1$. Inverting $g = 2D_N = \frac{6}{\pi^2}N^2 + O(N\log N)$ gives $N(g) \ge (\frac{\pi}{\sqrt{6}} - o(1))\sqrt{g}$. □

In moderate genera, omitting small quadratic cyclotomic factors ($d \in \{3,4,6\}$, each with $\varphi(d) = 2$) from the polynomial divisor of $h_B - h_A$ and instead imposing $F_3(B) = F_3(A)$, $F_4(B) = F_4(A)$, $F_6(B) = F_6(A)$ as affine-linear hyperplanes on the coefficients of h_A (solved via Hermite normal form and LLL lattice reduction around $C_g(X,q)$) yields examples in smaller genera:

Theorem 4.2 (Explicit higher-genus ordinary families)**.** *Explicit ordinary $\mathbf{Z}[q]$-families (h_A, h_B) establish the following lower bounds:*

g	$h_B(X) - h_A(X)$	*Matched counts*	*Threshold q_0*	*Lower bound*
15	$24q\,\Psi_1\Psi_2\Psi_5$	$c_1,\dots,c_6$ $(c_7(A) \ne c_7(B))$	1,086,853	$N(15) \ge 7$
24	$2\,\Psi_1\Psi_2\Psi_5\Psi_7$	$c_1,\dots,c_7$ $(c_8(A) \ne c_8(B))$	3,618,624,025	$N(24) \ge 8$
32	$2\,\Psi_1\Psi_2\Psi_5\Psi_7\Psi_8$	$c_1,\dots,c_8$ $(c_9(A) \ne c_9(B))$	11,921,384,751,788,836	$N(32) \ge 9$

In each case, for every prime power $q \ge q_0$ there exist non-isogenous g-dimensional ordinary abelian varieties $A, B/\mathbf{F}_q$ sharing $c_1,\dots,c_N$ and differing at c_{N+1}.

Proof. The integer coefficient arrays of $h_A, h_B \in \mathbf{Z}[q][X]$ (with maximum coefficient magnitudes 5,510, 30,861, and 1,136,960, respectively) and $g+1$ rational separating nodes $-2 \le x_0 < \cdots < x_g \le 2$ are recorded in the ancillary data file. For each $g \in \{15, 24, 32\}$:

(1) Every coefficient of X^{g-k} in h_A and h_B has q-degree at most $\lfloor k/2 \rfloor$, and $h_A(0) \equiv \pm 1 \pmod q$, $h_B(0) \equiv \pm 1 \pmod q$, giving ordinariness in every characteristic.

(2) Polynomial resultant evaluation over $\mathbf{Z}[q]$ confirms $\mathrm{Res}_X(\Psi_d, h_B) = \mathrm{Res}_X(\Psi_d, h_A)$ for $1 \le d \le N$ and $\mathrm{Res}_X(\Psi_{N+1}, h_B) - \mathrm{Res}_X(\Psi_{N+1}, h_A) \ne 0$ with no real roots on $[q_0, \infty)$.

(3) Writing $u_s = q^{-1/2}$, the scaled polynomials $H_A(x_m, u_s)$ and $H_B(x_m, u_s)$ alternate strictly in sign at $u_s = 0$ and have no roots on $(0, 1/K]$ (where $(K-1)^2 < q_0 \le K^2$), while every prime power $q \in [q_0, K^2)$ is checked via Sturm root counting on $(-2\sqrt{q}, 2\sqrt{q})$. Honda–Tate theory [Tat71, How95] completes the proof. □

Remark 4.3 (Asymptotic upper bounds on $N(g)$). While Theorem 4.1 establishes $N(g) = \Omega(\sqrt{g})$ and Theorems 2.1–2.3 show $N(g) < g$ for $g \in \{5, 6, 7\}$, the asymptotic upper-bound half of [CK26, Question 7.9]—in particular, whether $\limsup_{g\to\infty} N(g)/g < 1$ or $N(g) = o(g)$—remains open.

TABLE 1. Summary of known values and bounds for $N(g)$.

Genus g	$N(g)$	Effective threshold	Reference
1	1	all prime powers q	[CK26, Section 7]
2	2	odd q (lower), all q (upper)	[CK26, Section 7]
3	2	odd q (lower), $q \geq 16$ (upper)	[CK26, Section 7]
4	4	$q \geq 8$ (lower)	[CK26, Section 7]
5	4	$q \geq 8$ (lower), $q \geq 10^5$ (upper)	Theorem 2.1
6	4	$q \geq 8$ (lower), $q \geq 10^6$ (upper)	Theorem 2.2
7	5	$q \geq 5$ (lower), $q \geq 10^6$ (upper)	Theorem 2.3
8	$\in \{5, 6, 7, 8\}$	$q \geq 4$ (lower); no $\mathbf{Z}[q]$-family at $N = 5$	Theorem 3.1, Prop. 3.2
15, 24, 32	≥ 7, ≥ 8, ≥ 9	explicit ordinary $\mathbf{Z}[q]$-families	Theorem 4.2
$g \to \infty$	$\geq (\frac{\pi}{\sqrt{6}} - o(1))\sqrt{g}$	Chebyshev–totient families	Theorem 4.1

5. A Polynomial Threshold $Q(g) = (10g)^6$ for Zeta Recovery

In [CK26, Theorem 1.1], Chidambaram and Keller prove that $c_1(A), \ldots, c_g(A)$ determine the zeta function of A whenever $q > Q(g) = (16g^3 p(2g))^{2g+2}$ (with the inductive step of [CK26, Lemma 5.2] also formalized in Lean 4 [CK26b]), and ask in [CK26, Remark 7.12] whether $Q(g)$ can be chosen polynomial in g. In the analysis of [CK26, Sections 5–6], super-polynomial dependence on g arises in two places:

(1) In [CK26, Lemma 5.2], after isolating $\tilde{s}_k - s_k$ ($k \geq i$) in Macdonald's partition expansion of e_{2g-m} and e_m, the cofactor sum over partitions $\lambda \vdash r$ ($r \in \{2g-m-k,\ m-k\}$) in $s_1, \ldots, s_{i-1}$ is bounded term-by-term via the triangle inequality $\sum_{\lambda\vdash r} \prod_\ell \frac{|s_\ell|^{m_\ell}}{m_\ell! \ell^{m_\ell}} \leq p(r)(2gq^{-1/2})^r$, introducing $p(2g)$ into $D = 16g^3 p(2g)$ (which [CK26, Remark 7.12] notes can be reduced to $D = O(g^2)$, giving $Q(g) = g^{O(g)}$, via the partition generating function).

(2) In [CK26, Lemma 5.3], the error envelope $B_j = gq^{-j/2-g/4} D^{j-g}$ carries a negative exponent $j - g < 0$ in the seed range $g/2 < j < g$; matching the initial logarithm truncation error $|\tilde{s}_j - s_j| \leq 4gq^{-j} \leq B_j$ at $j = \lfloor g/2 \rfloor + 1$ then requires a power of q to dominate $D^{\lfloor (g-1)/2 \rfloor}$, producing the exponent $2g + 2$ on D.

Both sources of super-polynomial growth can be avoided within the framework of [CK26, Sections 4–6]: keeping the signed cofactor sum over $\lambda \vdash r$ intact before taking absolute values—equivalently, expanding $e_M(\tilde{s}) - e_M(s)$ via Newton's exponential generating function—identifies the coefficient of $(\tilde{s}_k - s_k)/k$ directly as $(-1)^{k-1} e_r(s)$ (with $|e_r(s)| \leq \binom{2g}{r} q^{-r/2} \leq \frac{(2g)^r}{r!} q^{-r/2}$), and re-centering the geometric envelope at $(g+1)/2$ rather than g keeps the exponent $j - (g+1)/2$ non-negative across the seed range $j > g/2$. Crucially, at $D = E$ the modified envelope B_j^* differs from B_j by the j-independent scalar $B_j^*/B_j = 4q^{-1/4} E^{(g-1)/2}$, so the homogeneous degree-1 linear contraction of [CK26, Lemma 5.2] is preserved automatically while transferring the $E^{\Theta(g)}$ amplification from the $q^{1/4}$-constrained seed index $j = \lfloor g/2 \rfloor + 1$ to the final rounding step $j = 2g$, where the $q^{(g+1)/4}$ margin absorbs $E^{(3g-1)/2}$ under a fixed polynomial condition $q > E^6$.

Theorem 5.1 (Polynomial threshold $Q(g) = (10g)^6$, addressing [CK26, Remark 7.12])**.** *For every integer $g \geq 2$ and every prime power $q > (10g)^6 = 10^6 g^6$, the first g point counts $c_1(A), \ldots, c_g(A)$*

uniquely determine the zeta function of any g-dimensional abelian variety $A/\mathbf{F}_q$. Moreover, the recovery algorithm of [CK26, Sections 4–6] *recovers $f_A(T)$ without modification for all $q > (10g)^6$.*

Proof. Following [CK26, Section 2], let $f_A(T) = T^{2g}P_A(1/T) = \prod_{\ell=1}^{2g}(1-\alpha_\ell T)$ be the reciprocal Frobenius polynomial of A, set $\beta_\ell := \alpha_\ell^{-1}$ ($|\beta_\ell| = q^{-1/2}$), and write

$$f_A(T/q) = q^{-g}P_A(T) = \prod_{\ell=1}^{2g}(1-\beta_\ell T) = \sum_{r=0}^{2g}(-1)^r e_r T^r,$$

with power sums $s_m := \sum_{\ell=1}^{2g}\beta_\ell^m$ ($m \geq 1$, satisfying $q^m s_m \in \mathbf{Z}$ and $|s_m| \leq 2gq^{-m/2}$) and elementary symmetric polynomials $e_r = e_r(s)$ satisfying $|e_r(s)| \leq \binom{2g}{r}q^{-r/2} \leq \frac{(2g)^r}{r!}q^{-r/2}$ and the functional equation $e_{g+n}(s) = q^{-n}e_{g-n}(s)$ for $1 \leq n \leq g$ [CK26, Lemma 2.2].

Let $i \in \{1, \dots, g\}$ be the step index of [CK26, Sections 4–6], assuming $s_1, \dots, s_{i-1}$ have already been recovered exactly:

- *Step 1 ($1 \leq i \leq g/2$,* [CK26, Lemma 4.1]*):* This case requires only $q > (16g)^2 = 256g^2$, which holds for all $g \geq 2$ since $(10g)^6 \geq 20^6 = 6.4 \times 10^7 > 256g^2$ (as $10^6 g^4 > 256$).

Fix $g/2 < i \leq g$ and assume $q > E^6$, where $E := 10g \geq 20$. For any $m \in \{2i-1, 2i\}$ with $m > g$, suppose approximations $\tilde{s}_1, \dots, \tilde{s}_{m-1}$ satisfy $\tilde{s}_k = s_k$ for $1 \leq k \leq i-1$, and write $\delta_k := \tilde{s}_k - s_k$ for $i \leq k \leq m-1$. In the formal power series ring $\mathbf{R}[[T]]$, Newton's exponential identity $\sum_{r\geq 0}(-1)^r e_r(s)T^r = \exp(-\sum_{k\geq 1}\frac{s_k}{k}T^k)$ gives, modulo T^{m+1},

$$\sum_{r=0}^{m}(-1)^r e_r(\tilde{s}_{<m}, s_m)T^r \equiv \left(\sum_{r\geq 0}(-1)^r e_r(s)T^r\right)\exp\left(-\sum_{k=i}^{m-1}\frac{\delta_k}{k}T^k\right) \pmod{T^{m+1}}.$$

Since $2i > m-1$ for $m = 2i-1$, and $2i = m$ is the unique quadratic index $\leq m$ when $m = 2i$, while $2g - m < g < 2i$, extracting coefficients of T^{2g-m} and T^m yields the identities

$$e_{2g-m}(\tilde{s}) - e_{2g-m}(s) = \sum_{k=i}^{2g-m}(-1)^{k-1}\frac{\delta_k}{k}\,e_{2g-m-k}(s), \tag{6}$$

$$e_m(\tilde{s}_{<m}, s_m) - e_m(s) = \sum_{k=i}^{m-1}(-1)^{k-1}\frac{\delta_k}{k}\,e_{m-k}(s) + \mathbf{1}_{m=2i}\,\frac{\delta_i^2}{2i^2}, \tag{7}$$

where $e_m(\tilde{s}_{<m}, s_m) - (-1)^{m-1}\frac{s_m}{m}$ is the polynomial in $(\tilde{s}_1, \dots, \tilde{s}_{m-1})$ used in [CK26, Lemma 5.1] to compute $\tilde{s}_m$. Substituting (6)–(7) into the formula of [CK26, Lemma 5.1] for $\frac{1}{m}(\tilde{s}_m - s_m)$ gives

$$|\tilde{s}_m - s_m| \leq m\sum_{k=i}^{2g-m}\frac{|\delta_k|}{k}\,q^{-(m-g)}|e_{2g-m-k}(s)| + m\sum_{k=i}^{m-1}\frac{|\delta_k|}{k}\,|e_{m-k}(s)| + \mathbf{1}_{m=2i}\,\frac{|\delta_i|^2}{i}. \tag{8}$$

Define the modified error envelope for all $j \geq (g+1)/2$:

$$B_j^* := 4g\,q^{-j/2-(g+1)/4}\,E^{j-(g+1)/2}, \qquad E = 10g.$$

We check the base case, inductive propagation, and final integer rounding:

(1) *Base case ($g/2 < j \leq g$, cf.* [CK26, Lemma 5.3]*):* Since j is an integer strictly greater than $g/2$, we have $2j \geq g+1$, so $j - (g+1)/2 \geq 0$. Because $q > 1$ and $E \geq 20 > 1$, the initial logarithm truncation error $|\tilde{s}_j - s_j| \leq 4gq^{-j}$ of [CK26, Lemma 5.3] satisfies

$$\frac{4gq^{-j}}{B_j^*} = \left(q^{1/2}E\right)^{-(j-(g+1)/2)} \leq 1, \qquad \text{hence} \qquad |\tilde{s}_j - s_j| \leq B_j^*.$$

(2) *Inductive step ($g/2 < i \leq g$, refining* [CK26, Lemma 5.2]*):* Suppose $|\delta_k| \leq B_k^*$ holds for $i \leq k \leq 2i-2$. We first establish $|\delta_m| \leq B_m^*$ for $m = 2i-1$ (if $2i-1 = g$, this already holds by the base case; otherwise $2i-1 > g$), and then for $m = 2i$. Setting $r := m-k \in \{1, \dots, m-i\}$,

note that $r \le i \le g$, $m/k \le 2i/i = 2$, $|e_r(s)| \le \frac{(2g)^r}{r!} q^{-r/2}$, and in the first sum of (8) (where $k \le 2g - m$, i.e. $r' \coloneqq 2g - m - k = r - 2(m-g) \ge 0$),

$$q^{-(m-g)}|e_{r'}(s)| \le q^{-(m-g)} \binom{2g}{r'} q^{-r'/2} \le \binom{2g}{r} q^{-r/2} \le \frac{(2g)^r}{r!} q^{-r/2}$$

since $0 \le r' < r \le g$ implies $\binom{2g}{r'} \le \binom{2g}{r}$. Moreover, $B^*_{m-r} q^{-r/2} = B^*_m E^{-r}$. Thus the two linear sums in (8) together contribute at most

$$2 \sum_{r=1}^{m-i} 2B^*_{m-r} \frac{(2g)^r}{r!} q^{-r/2} = 4B^*_m \sum_{r=1}^{m-i} \frac{(2g/E)^r}{r!} < 4(e^{1/5} - 1) B^*_m < 0.88562\, B^*_m.$$

When $m = 2i$, the quadratic term $\frac{|\delta_i|^2}{i} \le \frac{(B^*_i)^2}{i}$ satisfies, since $i > g/2$ and $q > E^6 \ge 20^6$,

$$\frac{(B^*_i)^2/i}{B^*_{2i}} = \frac{4g}{i}\, q^{-(g+1)/4}\, E^{-(g+1)/2} < 8\, E^{-2(g+1)} \le 8 \cdot 20^{-6} = 1.25 \times 10^{-7}.$$

Hence $|\tilde{s}_m - s_m| < (0.88562 + 1.25 \times 10^{-7}) B^*_m < 0.89\, B^*_m < B^*_m$, completing the induction up to $m = 2i$.

(3) *Final integer rounding ($g/2 < i \le g$, following* [CK26, Section 6]*):* As in [CK26, Section 6], s_i is recovered from $c_i(A)$ and $\tilde{s}_{2i}$ up to error $|\hat{s}_i - s_i| \le \frac{1}{2}|\tilde{s}_{2i} - s_{2i}| + 4gq^{-3i/2} \le \frac{1}{2}B^*_{2i} + 4gq^{-3i/2}$, and rounding $q^i \hat{s}_i$ to the nearest integer recovers $q^i s_i \in \mathbf{Z}$ uniquely provided $\frac{1}{2}B^*_{2i} < \frac{1}{4}q^{-i}$ and $4gq^{-3i/2} < \frac{1}{4}q^{-i}$. Since $i > g/2 \ge 1$ implies $i \ge 2$ and $q > (10g)^6 > 256g^2$, we have $q^{i/2} \ge q > 16g$, giving $4gq^{-3i/2} < \frac{1}{4}q^{-i}$. Finally, $\frac{1}{2}B^*_{2i} < \frac{1}{4}q^{-i}$ is equivalent to $q^{(g+1)/4} > 8g\, E^{2i-(g+1)/2}$, which is strictest at $i = g$: raising both sides to the 4-th power gives

$$\left(8g\, E^{2g-(g+1)/2}\right)^4 = (8g)^4 (10g)^{6g-2} = \left((10g)^6\right)^{g+1} \left(\frac{8g}{(10g)^2}\right)^4 = \left((10g)^6\right)^{g+1} \left(\frac{2}{25g}\right)^4 < q^{g+1}$$

for every $g \ge 2$ and $q > (10g)^6$. Thus $|\hat{s}_i - s_i| < \frac{1}{2}q^{-i}$, recovering s_i (and hence $s_1, \dots, s_g$ and $f_A(T)$) uniquely. □

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